Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Session	**2025-26 (EVEN)**	**Course Name**	**AIML Lab**
Semester	**6**	**Course Code**	**23CT1602**
Roll No	**23**	**Name of Student**	**Srushti Chaudhari**

Practical Number	5
Course Outcome	CO1: Apply Artificial Intelligence problem formulation techniques, intelligent agents, search algorithms, and logical inference to solve real-world problems. CO2: Implement and evaluate machine learning algorithms including classification, clustering, and dimensionality reduction using appropriate preprocessing and performance metrics. CO3: Analyze and compare ensemble learning models.
Aim	Classification using K-Nearest Neighbors (KNN) and Bayes Classifier To implement **K-Nearest Neighbors (KNN)** and **Bayes Classifier** for classification and evaluate the model performance using metrics such as **accuracy, precision, and recall**.
Problem Definition	Classification is a supervised learning technique used to predict the category or class label of given data. In this practical, we implement **K-Nearest Neighbors (KNN)** and **Naïve Bayes classifier** to classify data from a dataset such as the **Iris dataset or student performance dataset**. Before applying the model, the dataset is preprocessed by handling missing values and scaling features. The KNN algorithm classifies a new data point based on the majority class among its **k nearest neighbors**. The value of **k** is tuned to improve the model performance. The model is then evaluated using **accuracy, precision, and recall metrics**.
Theory (100 words)	K-Nearest Neighbors (KNN) is a supervised machine learning algorithm used for classification and regression tasks. It works by finding the **k nearest data points** to a new input based on distance measures such as Euclidean distance. The class label is assigned based on the **majority vote** of the nearest neighbors. Naïve Bayes is another classification algorithm based on **Bayes' theorem**, which assumes independence between features. It calculates the probability of each class given the input features and selects the class with the highest probability. Both algorithms are widely used for classification problems and are evaluated using metrics such as accuracy, precision, and recall.

Procedure and Execution

(100 Words)

Algorithm:

1. **Start the program.**
2. **Import necessary libraries.**
3. **Load the dataset (Iris dataset).**
4. **Perform data preprocessing such as feature selection and splitting the dataset into training and testing sets.**
5. **Train the KNN classifier with a selected value of k.**
6. **Tune the value of k to improve model performance.**
7. **Predict the class labels for test data.**
8. **Evaluate the model using accuracy, precision, and recall.**
9. **Display the results.**
10. **Stop the program.**

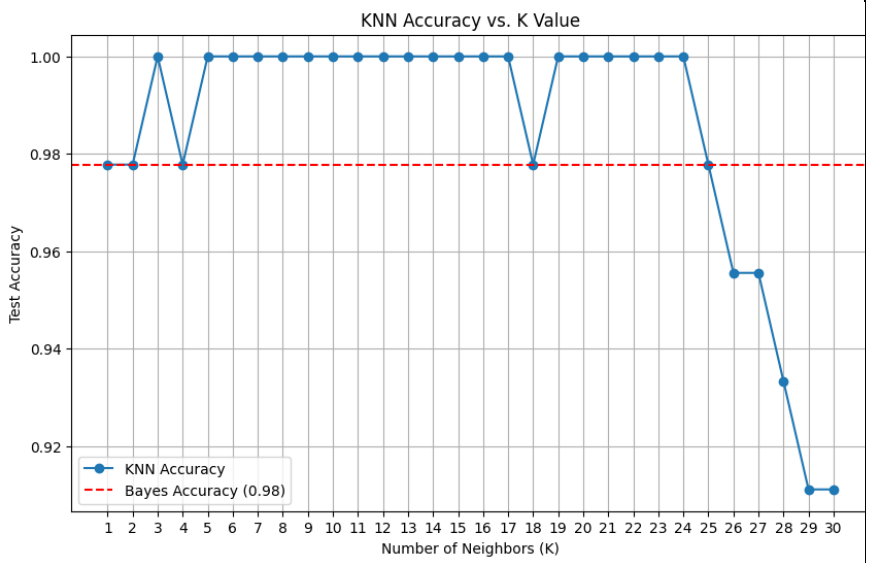
Code:

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import accuracy_score, precision_score,
recall_score

# Load dataset
data = load_iris()
X = data.data
y = data.target

# Train-test split
X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.2)

# KNN Model
knn = KNeighborsClassifier(n_neighbors=3)
knn.fit(X_train, y_train)

y_pred_knn = knn.predict(X_test)

print("KNN Results")
print("Accuracy:", accuracy_score(y_test, y_pred_knn))
print("Precision:", precision_score(y_test, y_pred_knn,
average='macro'))
print("Recall:", recall_score(y_test, y_pred_knn, average='macro'))

# Naive Bayes Model
nb = GaussianNB()
nb.fit(X_train, y_train)

y_pred_nb = nb.predict(X_test)

print("\nNaive Bayes Results")
print("Accuracy:", accuracy_score(y_test, y_pred_nb))
print("Precision:", precision_score(y_test, y_pred_nb,
```

```
average='macro'))
print("Recall:", recall_score(y_test, y_pred_nb, average='macro'))
```

Output:

```
•••  KNN Results
     Accuracy: 0.9333333333333333
     Precision: 0.9333333333333332
     Recall: 0.9333333333333332

     Naive Bayes Results
     Accuracy: 0.9333333333333333
     Precision: 0.9333333333333332
     Recall: 0.9333333333333332
```

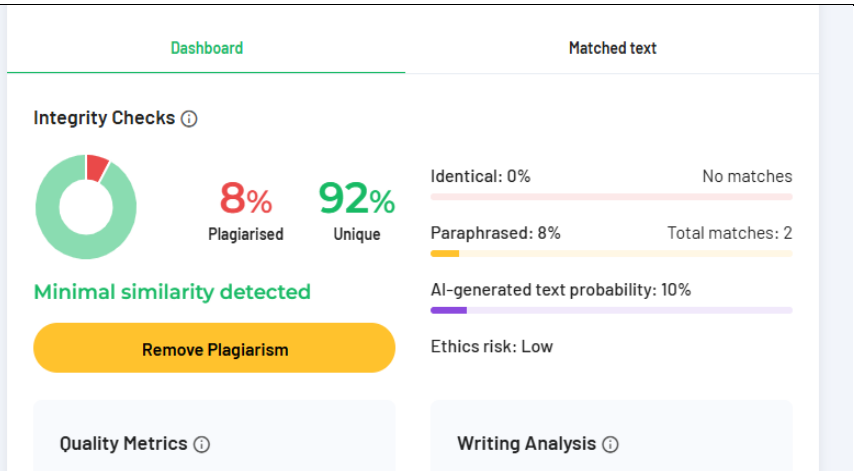


Optimal K: 3 with accuracy 1.0000
Bayes Classifier accuracy: 0.9778

Output Analysis

Comparison and Analysis
KNN classifies data based on similarity with nearby data points, while Naïve Bayes calculates probabilities using Bayes' theorem. KNN works well when the dataset is small and features are meaningful. Naïve Bayes works efficiently for large datasets and assumes feature independence. Both models can be evaluated using accuracy, precision, and recall to determine which performs better.

Output Analysis
The results show the classification performance of the KNN and Naïve Bayes algorithms. The accuracy, precision, and recall values indicate how well the model predicts the correct class labels. By adjusting the value of k in the KNN algorithm, the performance can be improved.

Link of student Github profile where lab assignment has	https://github.com/SRchaudhari15/AIML-Practicals.git
Conclusion	The KNN and Naïve Bayes classification models were successfully implemented using the Iris dataset. Data preprocessing and model training were performed, and the models were evaluated using accuracy, precision, and recall. The experiment demonstrates how classification algorithms can be applied to predict class labels in supervised machine learning problems.
Plag Report (Similarity index < 12%)	
Date	16/02/26

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